US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12416222
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Nair; Sajith et al.

Non-metallic subsea skid apparatus and methods

Abstract

A subsea skid (**100**) includes a plurality of skid tubes (**101-112**), and a plurality of skid joints (**130-135**) coupled between the plurality of skid tubes (**101-112**). The plurality of skid tubes (**101-112**) and the plurality of skid joints (**130-135**) form a frame. The plurality of skid tubes (**101-112**) and the plurality of skid joints (**130-135**) are each formed of a non-metallic material.

Inventors: **Nair; Sajith (Dubai, AE), Kulshrestha; Manoj (Dubai, AE), Singh; Nawin (Dubai, AE), Waimer; Johannes (Dubai, AE), Modica; Antonino Ascone (Dubai, AE), Shendre; Amit Balkrishna (Abu Dhabi, AE)**

Applicant: **J. RAY MCDERMOTT, S.A.** (Houston, TX)

Family ID: **1000008818313**

Assignee: **J. Ray McDermott, S.A. (Houston, TX)**

Appl. No.: **17/918150**

Filed (or PCT Filed): **April 15, 2021**

PCT No.: **PCT/US2021/027465**

PCT Pub. No.: **WO2021/211830**

PCT Pub. Date: October 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230135913 A1	May. 04, 2023

Related U.S. Application Data

us-provisional-application US 63010369 20200415

Publication Classification

Int. Cl.: **E21B41/08** (20060101); **B29C70/06** (20060101); **F16L1/20** (20060101)

U.S. Cl.:

CPC **E21B41/08** (20130101); **B29C70/06** (20130101); **F16L1/20** (20130101);

Field of Classification Search

CPC: F16L (1/20); E21B (41/08); B29C (70/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3375856	12/1967	Howard	248/68.1	F16L 1/16
4784527	12/1987	Hunter et al.	N/A	N/A
8397463	12/2012	Allred, III	52/696	F16B 11/008
8528291	12/2012	Allred, III	52/81.3	E04C 3/08
2006/0201679	12/2005	Williams	N/A	N/A
2010/0192506	12/2009	Allred, III	403/205	F16B 11/008
2015/0337988	12/2014	Huang	405/184.4	F16L 3/08
2019/0323325	12/2018	Hestetun	N/A	E21B 41/08
2019/0374976	12/2018	Herrington	N/A	N/A
2019/0376250	12/2018	Grytdal et al.	N/A	N/A
2020/0103051	12/2019	Clunie	N/A	E21B 43/017
2021/0348703	12/2020	Vorley	N/A	B23P 19/061

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110884170	12/2019	CN	B29C 33/02
3590692	12/2019	EP	N/A
2355034	12/2000	GB	E02B 17/02
2440808	12/2007	GB	N/A
2577812	12/2019	GB	N/A
2006289857	12/2005	JP	B29C 65/00
WO-2016133393	12/2015	WO	B29C 65/485
2020070476	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion dated Jun. 21, 2021 for Application No. PCT/US2021/027465. cited by applicant
Saudia Arabia Examination Report issued to Patent Application No. 522440918 on Sep. 19, 2023. cited by applicant
A.R.I.P.O. Office Action issued to patent application No. AP/P/2022/014458 on Aug. 16, 2024. cited by applicant
United Arab Emirates Office Action and Search Report dated Oct. 6, 2024 for Application No. P6002128/2022. cited by applicant
Brazil Search Report issued to application No. BR112022020815-4 on Nov. 26, 2024. cited by applicant
Brazil Office Action issued to patent application No. BR112022020815-4 on May 6, 2025. cited by applicant
Notification of Non-Compliance With Substantive Requirements and Invitation To Submit Observations And/Or Amended Application Issued To Patent Application No. AP/P/2022/014458 on May 14, 2025. cited by applicant

---

Primary Examiner: Anderson; Amber R

Assistant Examiner: Lawson; Stacy N

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

BACKGROUND
Field
(1) Aspects of the present disclosure relate to non-metallic subsea skid apparatus and associated methods, such as composite subsea valve skid apparatus and methods.
Description of the Related Art
(2) Subsea skids can be affected by subsea conditions. For example, subsea conditions can corrode the subsea skids, which can involve maintenance and/or reduced lifespans for the subsea skids. The subsea skids can be heavy and can also be difficult to access, such as by diving operations. Furthermore, subsea skids can involve carbon footprints and can become snagged on other subsea components.
(3) Therefore, there is a need for subsea skid apparatus and related methods that facilitate increased lifespans, corrosion resistance, reduced weight, anti-snagging capabilities, and reduced carbon footprints.
SUMMARY
(4) Implementations of the present disclosure relate to non-metallic subsea skid apparatus and associated methods, such as composite subsea valve skid apparatus and methods.
(5) In one implementation, a subsea skid includes a plurality of skid tubes, and a plurality of skid joints coupled between the plurality of skid tubes. The plurality of skid tubes and the plurality of skid joints form a frame. The plurality of skid tubes and the plurality of skid joints are each formed of a non-metallic material.
(6) In one implementation, a method of forming a subsea skid includes disposing portions of a plurality of skid tubes in legs of a plurality of skid joints to arrange the plurality of skid tubes and the plurality of skid joints as a frame. Each of the plurality of skid joints and the plurality of skid

tubes is formed of a non-metallic material. The method includes curing the portions of the plurality of skid tubes to the legs of the plurality of skid joints.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above-recited features of the disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. **1**A is a schematic isometric view of a subsea skid, according to one implementation.

(3) FIG. **1**B is an enlarged schematic view of the subsea skid shown in FIG. **1**A, according to one implementation.

(4) FIG. **1**C is a schematic top cross-sectional view of one of the first set of skid joints shown in FIG. **1**B, along Section **1**C-**1**C, according to one implementation.

(5) FIG. **1**D is a schematic isometric exploded view of one of the support joints shown in FIG. **1**A, according to one implementation.

(6) FIG. **1**E is a schematic isometric exploded view of one of the skid joints shown in FIG. **1**A, according to one implementation.

(7) FIG. **2** is a schematic side view of the first side of the subsea skid shown in FIG. **1**A, according to one implementation.

(8) FIG. **3** is a schematic side view of the second side of the subsea skid shown in FIG. **1**A, according to one implementation.

(9) FIG. **4** is a schematic top view of the subsea skid shown in FIG. **1**A, according to one implementation.

(10) FIG. **5** is a schematic isometric view of a tube portion, according to one implementation.

(11) FIG. **6** is a schematic isometric view of a valve string fastened to the subsea skid, according to one implementation.

(12) FIG. **7** is a schematic block diagram view of a method of forming a subsea skid, according to one implementation.

(13) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one implementation may be beneficially utilized on other implementations without specific recitation.

DETAILED DESCRIPTION

(14) Aspects of the present disclosure relate to non-metallic subsea skid apparatus and associated methods, such as composite subsea valve skid apparatus and methods.

(15) FIG. **1**A is a schematic isometric view of a subsea skid **100**, according to one implementation. The subsea skid **100** is a valve subsea skid. The subsea skid **100** is non-metallic. The subsea skid **100** includes a plurality of skid tubes **101-112** coupled together using a plurality of skid joints **130-135**. Each of the skid tubes **101-112** and the skid joints **130-135** has one or more outer surfaces that have a smooth contour and/or a tapered finish. The subsea skid **100** shown in FIG. **1**A includes fifty skid tubes **101-112** and twenty-two skid joints **130-135**. The subsea skid **100** includes a plurality of padeye plates **141**. Each of the padeye plates **141** is formed in a first set of skid joints **132**. The first set of skid joints **132** include four skid joints **132**. The plurality of padeye plates **141** are lift points for attaching lift devices, such as hoist lines or hooks of cranes, to lift the subsea skid **100**. The skid joints **130-135** and the skid tubes **101-112** form a frame **150** of the subsea skid **100**. The subsea

skid **100** includes a first side **151**, a second side **152**, a third side **153**, and a fourth side **154**.
(16) The subsea skid **100** tubes include vertical skid tubes **109**, horizontal skid tubes **101**, **104-107**, **110-112**, and angled skid tubes **102**, **103**, **108**. At least some of the angled skid tubes **102** are disposed in a horizontal plane (parallel to the X-Y plane), and at least some of the angled skid tubes **103**, **108** are disposed in vertical planes (parallel to the Y-Z plane for the angled skid tubes **103** and parallel to the X-Z plane for the angled tubes **108**). Each of the skid joints **130-135** includes a plurality of legs protruding outward from a center of the respective skid joint **130-135**. Each leg receives a portion (such as an end portion) of one of the skid tubes **101-112**. The plurality of legs includes a number of legs, and the number of legs is within a range of two to five. In one example, the skid joints **130-135** include T-joints having three legs. In the implementation shown in FIG. **1**A, each of the skid joints **130-134** includes four legs or five legs. The skid joints **130-135** and the skid tubes **101-112** are cured together to form the subsea skid **100**. The curing occurs after the portions (such as the end portions) of the skid tubes **101-112** are disposed within the respective legs of the skid joints **130-135**.
(17) The skid joints **130-135** include a plurality of support joints **135**. The support joints **135** are valve support joints. The subsea skid **100** shown includes three support joints **135**. The support joints **135** are similar to the skid joints **130-134**. The support joints **135** include two legs **136***a*, **136***b*. Each of the support joints **135** includes a support flange **137** formed therein for supporting equipment, such as valve equipment. Each support flange **137** includes one or more fastener openings **138** formed therein. Each support joint **135** includes flanges **139***a*, **139***b* extending between the respective legs **136***a*, **136***b* and the support flange **137**. Each support flange **137** is configured to couple to equipment using one or more fasteners extending through the one or more fastener openings **138**.
(18) Each of the skid tubes **101-112** and the skid joints **130-135** is formed of a non-metallic material. The support flanges **137** and the padeye plates **141** are formed of the non-metallic material. In one embodiment, which can be combined with other embodiments, the non-metallic material is polymeric and includes one or more polymers. In one embodiment, which can be combined with other embodiments, the non-metallic material is a composite material. The composite material is a carbon composite, such as a carbon fiber material. In one embodiment, which can be combined with other embodiments, the non-metallic material includes one or more of glass fiber reinforced polymer (GFRP) and/or carbon fiber reinforced polymer (CFRP). Each of the skid tubes **101-112** and the skid joints **130-135** (such as the legs of the skid joints **130-135**) includes a tube wall thickness T**1** (shown in FIG. **5**). The tube wall thickness T**1** is within a range of 8 mm to 12 mm.
(19) The skid tubes **101-112** and the skid joints **130-135** each include one or more tube portions that are formed of the non-metallic material. The skid tubes **101-112** each include a single tube portion. The skid joints **130-135** each include a number of tube portions that is equal to the number of legs. In one embodiment, which can be combined with other embodiments, the tube portions are formed of filament wound carbon fiber tubes.
(20) FIG. **1**B is an enlarged schematic view of the subsea skid **100** shown in FIG. **1**A, according to one implementation. Each of the padeye plates **141** is formed on an outer surface of one of the first set of skid joints **132**. Each of the padeye plates **141** includes one or more lift openings **142** formed therein. A shackle, including a shackle body **193** and a shackle pin **194**, is coupled to each of the padeye plates **141**. Each shackle body **193** is configured to attach to one or more lift devices used to lift the subsea skid **100**.
(21) Each of the first set of skid joints **132** includes five legs **171-175**. An end portion **181-185** of each one of the respective skid tubes **103**, **108-111** is disposed in a respective leg **171-175** of the respective skid joint **132**. The respective end portion **181-185** is disposed in a central opening (e.g., a circular opening) of the respective leg **171-175**.
(22) FIG. **1**C is a schematic top cross-sectional view of one of the first set of skid joints **132** shown

in FIG. **1**B, along Section **1**C-**1**C, according to one implementation. A circumferential outer surface of each respective end portion **181-185** is cured to a circumferential inner surface of the respective leg **171-175**. A circumferential outer surface **186** is cured to a circumferential inner surface **187** of leg **174**. A circumferential outer surface **188** is cured to a circumferential inner surface **189** of leg **175**.

(23) Each of the skid tubes **101-112** and the skid joints **130-135** includes a plurality of sections cured together. Each of the plurality of sections is formed of the non-metallic material. Each section of the plurality of sections includes one or more curved portions and two or more planar flange portions extending relative to the one or more curved portions.

(24) FIG. **1**D is a schematic isometric exploded view of one of the support joints **135** shown in FIG. **1**A, according to one implementation. The support joint **135** includes a plurality of sections **191**, **192** (two are shown) cured together to form the support joint **135**. Each section **191**, **192** includes one or more curved portions **193-195** (three are shown) and two or more planar flange portions **196-198** (three are shown) extending relative to the one or more curved portions **193-195**. Two or more planar surfaces **199***a*-**199***d* (four are shown) of the two or more planar flange portions **196-198** of the section **192** are interfaced with and cured to two or more planar surfaces **199***a*-**199***d* of the two or more planar flange portions **196-198** of the section **191**.

(25) The skid tubes **101-112** can extend partially into respective legs of the skid joints **130-135** (as is shown in FIG. **1**C) or the skid tubes **101-112** can extend through and past respective legs of the skid joints **130-135** (as is shown in FIG. **1**D for the skid tube **105** extending through and past the curved portions **193**, **194** of each of the sections **191**, **192**). The curved portions **193**, **194** of the sections **191**, **192**, when cured together, form the two respective legs **136***a*, **136***b* (shown in FIG. **1**A) of the support joint **135**. In the implementation shown in FIG. **1**D, an intermediate portion of the skid tube **105** is disposed in the two respective legs **136***a*, **136***b* of the support joint **135**, and the intermediate portion is cured to the two respective legs **136***a*, **136***b* of the support joint **135**. The planar flange portions **196**, **197** of the sections **191**, **192**, when cured together, form the flanges **139***a*, **139***b* (shown in FIG. **1**A) of the support joint **135**. The planar flange portions **198** of the sections **191**, **192**, when cured together, form the support flange **137** (shown in FIG. **1**A) of the support joint **135**.

(26) FIG. **1**E is a schematic isometric exploded view of one of the skid joints **132** shown in FIG. **1**A, according to one implementation. The skid joint **132** includes a plurality of sections **1001-1007** (seven are shown) cured together to form the skid joint **132**. Each section **1001-1007** includes one or more curved portions and two or more planar flange portions extending relative to the one or more curved portions. Two or more planar surfaces of the two or more planar flange portions of each section **1001-1007** are interfaced with and cured to two or more planar surfaces of the two or more planar flange portions one or more adjacent sections **1001-1007**. The curved portions and the planar flange portions of the sections **1001-1007** are cured together to form the legs **171-175** shown in FIG. **1**B. A planar surface **1010** of a planar flange portion **1011** of section **1001** is interfaced with and cured to a planar surface **1012** of a planar flange portion **1013** of section **1002** to form the padeye plate **141** shown in FIG. **1**B. The sections of the skid joints **130-135** (such as the sections **1001-1007**) are formed prior to being cured together to form the respective skid joints **130-135**. In one embodiment, which can be combined with other embodiments, the sections of the skid joints **130-135** are formed using molding, such as composite molding. Other methods of forming the sections of the skid joints **130-135** are contemplated.

(27) FIG. **2** is a schematic side view of the first side **151** of the subsea skid **100** shown in FIG. **1**A, according to one implementation. Each of the skid tubes **103** is disposed in a plane parallel to the Y-Z plane at an oblique angle A**1** relative to an axis parallel to the Z-axis. The oblique angle A**1** is within a range of 30 degrees to 50 degrees, such as 33 degrees. The subsea skid **100** has a length L**1** that is within a range of 450 inches to 500 inches. The frame **150** includes (from the side view of the first side **151**) a trapezoidal pattern **162**. The frame **150** includes the trapezoidal pattern **162**

from a side view of the third side **153**.

(28) FIG. **3** is a schematic side view of the second side **152** of the subsea skid **100** shown in FIG. **1**A, according to one implementation. The subsea skid **100** has a height H**1** that is within a range of 115 inches to 150 inches. The frame **150** includes (from a side view of the third side **153**) a trapezoidal pattern **163**. The frame **150** includes the trapezoidal pattern **163** from a side view of the fourth side **154**. The support joints **135** extend above central axes of the horizontal skid tubes **101**, **102**, **104-107**, **112** by a height H**2** that is within a range of 20 inches to 30 inches. Each of the skid tubes **103** is disposed in a plane parallel to the X-Z plane at an oblique angle A**2** relative to an axis parallel to the Z-axis. The oblique angle A**2** is within a range of 30 degrees to 50 degrees, such as 33 degrees.

(29) FIG. **4** is a schematic top view of the subsea skid **100** shown in FIG. **1**A, according to one implementation. The subsea skid **100** has a width W**1** that is within a range of 250 inches to 275 inches. The frame **150** of the subsea skid **100** includes (from the top view) a rectangular pattern **157** having four sides and four corners, and two trapezoidal patterns **158***a*, **158***b* disposed outwardly of two opposing sides of the rectangular pattern **157**. The padeye plates **141** are formed on four skid joints **132** that are disposed respectively at the four corners of the rectangular pattern **157**. Each of the four padeye plates **141** extends inwardly toward a center C**1** of the rectangular pattern **157** from the respective corner of the four corners. Each of the skid tubes **102** is disposed in a plane parallel to the X-Y plane at an oblique angle A**3** relative to an axis parallel to the Y-axis. The oblique angle A**3** is within a range of 30 degrees to 50 degrees, such as 33 degrees.

(30) FIG. **5** is a schematic isometric view of a tube portion **500**, according to one implementation. The tube portion **500** can be used as the skid tubes **101-112**. The present disclosure contemplates that the tube portion **500** can be used as to form tube portions of the skid joints **130-135**, such as to form legs of the skid joints **130-135**. The tube portion **500** is a filament wound carbon fiber tube formed of filament wound carbon fiber. The filament wound carbon fiber is wound using a layup of plies. In one embodiment, which can be combined with other embodiments, the layup of plies is wound at an angle within a range of −45 degrees to 45 degrees. In one embodiment, which can be combined with other embodiments, the layup of plies is wound at an angle within a range of −10 degrees to 10 degrees. In one example, which can be combined with other examples, the filament wound carbon fiber includes a flexural modulus that is within a range of 70 GPa to 100 GPa, such as about 80 GPa.

(31) FIG. **6** is a schematic isometric view of a valve string **601** fastened to the subsea skid **100**, according to one implementation. The valve string **601** includes a plurality of valves **615**, **616** coupled to pipes **617**, **618**, **619**, such as steel pipes. The valve string **601** includes a plurality of flanges **610** fastened to the support flanges **137** using a plurality of fasteners **620**.

(32) FIG. **7** is a schematic block diagram view of a method **700** of forming a subsea skid, according to one implementation. Operation **702** includes disposing portions of a plurality of skid tubes in legs of a plurality of skid joints to arrange the plurality of skid tubes and the plurality of skid joints as a frame. The portions are end portions and/or intermediate portions of the skid tubes. The disposing the portions of the plurality of skid tubes in legs of the plurality of skid joints can include abutting planar surfaces of sections of each skid joint against each other to form the legs of the respective skid joint. Each of the plurality of skid joints and the plurality of skid tubes is formed of a non-metallic material.

(33) Operation **704** includes curing the portions of the plurality of skid tubes to the legs of the plurality of skid joints. In one embodiment, which can be combined with other embodiments, the curing includes heating and pressurizing the plurality of skid tubes and the plurality of skid joints in an autoclave. In one embodiment, which can be combined with other embodiments, the plurality of skid tubes and the plurality of skid joints are heated to a cure temperature within a range of 60 degrees Celsius to 120 degrees Celsius. In one embodiment, which can be combined with other embodiments, the curing occurs at an ambient temperature, such as at an offshore site. The ambient

temperature is up to 45 degrees Celsius. The present disclosure contemplates that other temperatures, such as temperatures higher than the ambient temperature and the cure temperature, can be used for the curing. The curing can include using a bonding material, which can include resin, to bond the plurality of skid tubes and the plurality of skid joints together. The curing can include curing sections of each respective skid joint together. Other curing operations are contemplated.

(34) Operation **706** includes fastening a plurality of flanges of a valve string to a plurality of support flanges of the plurality of skid joints. The present disclosure contemplates that equipment other than valve strings can be fastened to the plurality of support flanges of the plurality of skid joints.

(35) Benefits of aspects of the present disclosure (such as the non-metallic material) include lifecycle cost savings of 25%-30% and more, longer operational lifespans for subsea skids, modularity and flexibility in design and application, reduced lead times, anti-snagging and avoiding anchor wire catching, reduced or eliminated corrosion of skids, eliminated need for cathodic protection, reduced maintenance, increased and easier (less obstructions for) access for diver operations and ROV operations, weight reductions of 70% or more for skids, and reduced carbon footprint in manufacturing and operating skids.

(36) As an example, using the non-metallic material facilitates reduced weight and eliminated need for cathodic protection that would involve heavy sacrificial anodes. The eliminated need for cathodic protection reduces or eliminates the need for maintenance conducted on the subsea skid **100**. Using the non-metallic material also facilitates the ability to use larger gaps between the skid tubes **101-112** (such as larger gaps in the rectangular pattern **157** and the trapezoidal patterns **158***a*, **158***b*) to reduce obstructions for access to components of the subsea skid **100**. As another example, the non-metallic material facilitates stress performance under loading applied as a result of operational conditions, such as water pressure at deep subsea levels and loading resulting from lifting and supporting the subsea skid **100** at the padeye plates **141** to lower the subsea skid **100** to the seafloor. The non-metallic material also facilitates stress performance when an object, such as other subsea equipment, contacts the subsea skid **100**. The non-metallic material, the skid joints, and the skid tubes facilitate modularity and flexibility in design and application of subsea skids.

(37) Lifting expenditures are reduced as the lifting forces needed to lift and manipulate the subsea skid **100** are reduced. The lower lifting expenditures facilitate use of smaller offshore vessels. Using the non-metallic material, smooth contours and tapered finishes can be used on outer surfaces of the subsea skid **100** to facilitate reduced wire catching and snagging, such as reduced anchor wire snagging. For example, the contours of the curved portions and the planar flange portions of the skid joints **130-135** facilitate reduced protrusions, which facilitates reduced wire catching and snagging.

(38) It is contemplated that one or more of the aspects disclosed herein may be combined. Moreover, it is contemplated that one or more of these aspects may include some or all of the aforementioned benefits. As an example, it is contemplated that one or more of the aspects, features, components, and/or properties of the subsea skid **100**, the tube portion **500**, the valve string **601**, and/or the method **700** can be combined.

(39) It will be appreciated by those skilled in the art that the preceding embodiments are exemplary and not limiting. It is intended that all modifications, permutations, enhancements, equivalents, and improvements thereto that are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the scope of the disclosure. It is therefore intended that the following appended claims may include all such modifications, permutations, enhancements, equivalents, and improvements. The present disclosure also contemplates that one or more aspects of the embodiments described herein may be substituted in for one or more of the other aspects described. The scope of the disclosure is determined by the claims that follow.

Claims

1. A subsea skid comprising: a plurality of skid tubes; and a plurality of skid joints coupled between the plurality of skid tubes, the plurality of skid tubes and the plurality of skid joints forming a frame, wherein the plurality of skid tubes and the plurality of skid joints are each formed of a non-metallic material, wherein each of the plurality of skid joints comprises two or more legs, and each of the two or more legs of each skid joint comprises a cure joint comprising bonding material disposed between a circumferential inner surface of each of the two or more legs and a circumferential outer surface located on an end portion of one of the plurality of skid tubes, wherein a plurality of padeye plates are formed on outer surfaces of the plurality of skid joints, and wherein the plurality of skid joints comprises a plurality of support joints, each support joint comprising a support flange formed therein, and further comprising a valve string comprising a plurality of flanges fastened to the support flange of each support joint of the plurality of support joints.
2. The subsea skid of claim 1, wherein each skid joint of the plurality of skid joints comprises a plurality of sections cured together to form the respective skid joint, and each section of the plurality of sections comprises one or more curved portions and two or more planar flange portions extending relative to the one or more curved portions.
3. The subsea skid of claim 1, wherein the plurality of skid tubes comprise a plurality of horizontal tubes, a plurality of vertical tubes, and a plurality of angled tubes.
4. The subsea skid of claim 3, wherein the frame comprises a rectangular pattern having four corners and four sides, and two trapezoidal patterns disposed outwardly of two opposing sides of the four sides of the rectangular pattern, and the plurality of padeye plates are formed on outer surfaces of four skid joints of the plurality of skid joints that are disposed, respectively, at the four corners of the rectangular pattern.
5. The subsea skid of claim 4, wherein each of the plurality of skid tubes and each of the plurality of skid joints comprises a tube wall thickness, and the tube wall thickness is within a range of 8 mm to 12 mm.
6. The subsea skid of claim 4, wherein each of the plurality of skid tubes and each of the plurality of skid joints comprises one or more tube portions, and the one or more tube portions comprise one or more filament wound carbon fiber tubes.
7. The subsea skid of claim 1, wherein the non-metallic material comprises carbon.
8. The subsea skid of claim 1, wherein the non-metallic material is a composite material.
9. The subsea skid of claim 1, wherein the non-metallic material comprises carbon fiber.
10. The subsea skid of claim 1, wherein the non-metallic material is polymeric.
11. The subsea skid of claim 1, wherein the non-metallic material comprises glass fiber reinforced polymer (GFRP) and carbon fiber reinforced polymer (CFRP).
12. A subsea skid comprising: a plurality of skid tubes comprising a plurality of horizontal tubes, a plurality of vertical tubes, and a plurality of angled tubes; and a plurality of skid joints coupled between the plurality of skid tubes, the plurality of skid tubes and the plurality of skid joints forming a frame, wherein the frame comprises a rectangular pattern having four corners and four sides, and two trapezoidal patterns disposed outwardly of two opposing sides of the four sides of the rectangular pattern, and a plurality of padeye plates are formed on outer surfaces of four skid joints of the plurality of skid joints that are disposed, respectively, at the four corners of the rectangular pattern, and wherein the plurality of skid tubes and the plurality of skid joints are each formed of a non-metallic material, wherein each of the plurality of skid joints comprises two or more legs, and each of the two or more legs of each skid joint comprises a cure joint comprising bonding material disposed between a circumferential inner surface of each of the two or more legs and a circumferential outer surface located on an end portion of one of the plurality of skid tubes,

wherein each of the plurality of skid tubes and each of the plurality of skid joints comprises one or more tube portions, and the one or more tube portions comprise one or more filament wound carbon fiber tubes, and wherein each of the one or more filament wound carbon fiber tubes comprises a layup of plies that is wound at an angle within a range of −45 degrees to 45 degrees.

13. The subsea skid of claim 12, wherein each of the one or more filament wound carbon fiber tubes comprises a layup of plies that is wound at an angle within a range of −10 degrees to 10 degrees.

14. A method of forming a subsea skid, the method comprising: disposing a circumferential outer surface located on an end portion of one skid tube of a plurality of skid tubes in a circumferential inner surface of a leg of a plurality of legs of a plurality of skid joints to arrange the plurality of skid tubes and the plurality of skid joints as a frame, wherein a bonding agent is disposed between the circumferential inner surface of each of the plurality of legs and the circumferential outer surface located on an end portion of one of the plurality of skid tubes, and wherein each of the plurality of skid joints and the plurality of skid tubes is formed of a non-metallic material; curing the bonding agent disposed between the circumferential inner surface of each of the plurality of legs and the circumferential outer surface located on an end portion of one of the plurality of skid tubes by heating the portions of the plurality of skid tubes and the legs of the plurality of skid joints to a temperature of 60° C. to 120° C.; and fastening a plurality of flanges of a valve string to a plurality of support flanges of the plurality of skid joints, wherein each of the plurality of skid tubes and each of the plurality of skid joints comprises one or more tube portions, and the one or more tube portions comprise one or more filament wound carbon fiber tubes.